

Increased Anal Cancer, Dysplasia and Anal HR-HPV Positivity in Inflammatory Bowel Disease

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Introduction: Patients with inflammatory bowel disease (IBD) have increased anal cancer rates. We evaluated whether IBD subtype, documented anal high-risk HPV positivity, or perianal disease was associated with higher observed frequencies of anal cancer/dysplasia.

Methods: We performed a retrospective Epic Cosmos analysis of 305,065,760 patients with active diagnoses from 6/1990 to 6/2026, stratified as Crohn's disease, ulcerative colitis, or non-IBD. Anal HR-HPV positivity, anal cancer/dysplasia, and perianal disease were defined by ICD-10 codes. Chi-square testing used $p < 0.05$. Although approximately 2.4 million IBD patients were identified, HPV testing denominators, negative results, screening frequency, and vaccination status were unavailable; findings may reflect screening, surveillance, or coding differences.

Results: Compared with non-IBD patients, anal cancer/dysplasia was 5.0-fold higher in Crohn's disease and 6.4-fold higher in ulcerative colitis. Anal HR-HPV positivity was 2.2-fold and 3.6-fold higher, respectively. Anal cancer/dysplasia within 5 years after anal HR-HPV diagnosis was 2.6-fold higher in Crohn's disease and 4.1-fold higher in ulcerative colitis. The largest relative increase followed perianal disease diagnosis: 14.3-fold higher in Crohn's disease and 12.3-fold higher in ulcerative colitis. Unadjusted odds ratios versus non-IBD were similar, with the largest associations after perianal disease diagnosis: OR 14.26 for Crohn's disease and OR 12.26 for ulcerative colitis.

Patient Type	Anal Cancer or Dysplasia (total)	Anal Cancer or Dysplasia 5 years after Perianal Disease diagnosis	Positive High-Risk HPV diagnosis in anal area (total)	Anal cancer or dysplasia 5 years after High-Risk HPV diagnosis
Non-IBD (n=305,065,760)	128,173 (0.0420%)*	10,195 (0.0033%)*	22,867 (0.0075%)*	9,404 (0.0031%)*
Crohn's Disease (n=1,206,873)	2,535 (0.2100%)*	575 (0.0476%)*	197 (0.0163%)*	97 (0.0080%)*
Ulcerative Colitis (n=1,191,972)	3,227 (0.2707%)*	488 (0.0409%)*	319 (0.0268%)*	149 (0.0125%)*

*=Row-level Pearson chi-square test $p < 0.001$.

Conclusion: Patients with IBD had higher observed frequencies of anal cancer/dysplasia and anal HR-HPV positivity than non-IBD patients. The strongest association was seen in Crohn's

disease after perianal disease diagnosis. These findings support further study of targeted anal cancer screening strategies in IBD patients with anal HR-HPV positivity or prior perianal disease.

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